

OXYPC INVENTORY LIFECYCLE

OPERATIONS

UAT & Disaster Recovery

How releases are tested and signed off, how to recover the app when something breaks, and how it is deployed and run.

Last updated: 6 Nov 2026 · Source: staging branch @ 55ce58a

App Development Team · Bluemonk

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PART A

UAT Plan & Sign-off

Purpose: one repeatable User Acceptance Test (UAT) cycle per release, run on **staging**, signed by a business owner for each of the 12 personas. This plan reuses the existing test assets. It does not replace them.

1. Existing assets this plan reuses

Asset	What it holds	How this plan uses it
`OxyPC_UAT_Scenarios_v1.0.docx` (Mar 2026)	18 modules, IDs `UAT-AUTH-001` … `UAT-E2E-003`, plus a UAT Sign-Off Sheet	Source of step-level scenarios. Section 5 maps each persona to these IDs.
`OxyPC_UAT_Test_Cases.xlsx`	~80 test cases in 16 module sheets (`TC-AUTH-001`, `TC-LOT-001`, `TC-IQC-001`, `TC-L1-001`, `TC-QC-001`, `TC-COS-001`, `TC-SALE-001`, `TC-RET-001`, `TC-PARTS-001`, `TC-RAM-001`, `TC-BULK-001`, `TC-MASTER-001`, `TC-ADMIN-001`, `TC-RPT-001`, `TC-SEC-001` …), Execution Tracker, Defect Log, Test Data	Execution tracker for each cycle. Copy per release, fill "Executed By / Date / Pass-Fail / Defect Ref".
`docs/OxyPC_UAT_TestPlan.xlsx` (27 Apr 2026)	Phase P1–P3 route checks, Security Tests, Showstoppers, All Auto Results (48 auto tests, generated by `run_uat_excel.py`)	Automated smoke on staging before manual UAT. Showstoppers re-checked (Roadmap N5).
QA Dashboard `/qa` (`routers/qa_uat.py`, `models/qa_uat.py`)	Requirements, Test Cases, Executions, Defects, UAT Scenarios (with `sign_off_by` / `sign_off_at`), Releases, RTM	System of record. Every defect found in UAT is logged at `/qa/defects`; every scenario result at `/qa/uat`; the release moves `In UAT → UAT Done → Approved` at `/qa/releases`.
`seed_uat_users.py`	One UAT login per built-in role (`uat_admin`, `uat_invmgr`, `uat_iqc`, `uat_l1`, `uat_l2`, `uat_l3`, `uat_qc`, `uat_sales`, `uat_spares`)	Test accounts. Add custom-role accounts (Production Manager, Cosmetic Lead, Final QC, Accounts, TRC Manager) before the cycle.
`e2e_uat_test.py`, `tests/` (173 test files, `pytest.ini`)	Automated workflow and unit tests	Must pass before UAT entry.

2. Scope

In scope	Out of scope
Device lifecycle: Lot → GRN → IQC → Stock In → Production → L1/L2, L3/L4 → Stress → Cosmetic (6 stages) → Final QC → Ready to Sale → Sold → Return / Credit Note / Replacement	CRM sourcing, telecalling, dealer calling (own UAT)
Parts: request, handover, verify, Not In Stock, Procure, PNA, Parts GRN (As New), Harvest, warranty part cycle	Reliance FieldOps (`/fieldops`) — own 147 checks
Sales: Ready to Sale pricing, block tags, sale, invoice print, delivery challan, gate pass, transport update, returns, credit note, extended warranty	Trade Partner portal
Accounts: customer receipts, supplier payments, sales list, daily stock	WhatsApp service
Management: Overall Dashboard, Business P&L, Lot P&L, Daily Stock, Stock Aging, Receivables, Overdue, Physical Audits, Tickets	Performance / load testing (separate)
Permissions: each persona sees only its pages; negative tests from `UAT-RBAC-001…006`	
Any PRD in the release (PRD-01…04) — scenarios added from its acceptance criteria	

3. Environments

Environment	Purpose	Data	Who deploys	Notes
Local / Dev	Developer checks (`verify-feature` , `regression-sweep` skills)	Seed data (`seed_dummy_data.py`)	Developer	Not used for sign-off
Staging	**UAT and sign-off**	Copy of production restored from the nightly backup (`deploy/oxypc-backup.sh`), customer names / phones / GSTIN masked	App Development Team	**To be provisioned** (Roadmap N1). Today `deploy/oxypc-autodeploy.sh` deploys `origin/main` straight to production; there is no staging host in the repo. Use the `staging` branch; record "Staging" in the QA Dashboard environment field.
Production	Live use	Real	Auto-deploy after approval	Deploy only after Exit criteria are met and release status is `Approved`

Staging set-up checklist:

#	Item	Done
1	Separate DB and `.env` (no production secrets)	☐
2	`db_validator.py` passes on start-up	☐
3	UAT users created (`seed_uat_users.py` + custom-role users)	☐
4	Diagnose / OxyQC agent points at staging, not production	☐
5	Outbound WhatsApp / email / partner webhooks disabled	☐
6	Banner or different sidebar colour so testers know it is staging	☐

4. Entry and exit criteria

Entry (all must be true before testers start)

#	Criterion	Evidence
E1	Release built from the `staging` branch and deployed to staging	Release row at `/qa/releases` with status `In UAT`
E2	`pytest` green on the release commit	CI / console log attached to the release
E3	Full template parse: 0 errors (`regression-sweep` Step 1)	Log
E4	Route scan at baseline (`regression-sweep` Step 2: exactly 9 known flagged routes). Note: the skill calls `_scan_routes.py`, which is not in this repo checkout — locate it before the cycle.	Log
E5	`run_uat_excel.py` auto-smoke run on staging; no new failures vs last run	`OxyPC_UAT_TestPlan.xlsx` copy
E6	No open **Critical** defects for the release at `/qa/defects`	Filter: severity = Critical, status ≠ Closed
E7	Permission matrix reviewed for each persona's role (custom roles included)	Screenshot of `/admin/master/permissions` per role
E8	Release notes list every change and the personas affected	Release description

Exit (all must be true to sign off)

#	Criterion
X1	100% of in-scope scenarios executed; ≥ 95% Pass
X2	0 open Critical and 0 open High defects
X3	Medium / Low defects open only with an agreed fix date and owner
X4	End-to-end journey E2E-01 passed with the same tag from Lot to Credit Note
X5	Dashboard and report figures reconcile for the test tags (see 05)
X6	Sign-off form (Section 9) signed by every persona owner in scope
X7	Release status moved to `UAT Done`, then `Approved` by Managers/Admin

5. Roles and sign-off owners

#	Persona	UAT login (built-in or to create)	Sign-off owner (business)	Existing scenario IDs
1	IQC Inspector	`uat_iqc`	IQC lead	UAT-IQC-001...005, TC-IQC-*
2	Inventory Manager	`uat_invmgr`	Inventory Manager	UAT-LOT-001...004, UAT-STK-001...002, UAT-TRF-001...004, UAT-DEV-001...005
3	Production Manager	create custom-role user	Production Manager	(new — Section 6.3)

#	Persona	UAT login (built-in or to create)	Sign-off owner (business)	Existing scenario IDs
4	Repair Engineer L1/L2, L3/L4	`uat_l1` , `uat_l3`	Repair supervisor	UAT-L1R-001…006, UAT-L3R-001…004 (UAT-L2R-* legacy)
5	QC Inspector (Stress, OS install, diagnose)	`uat_qc`	QC lead	UAT-QC-001…004, TC-QC-*
6	Cosmetic Lead	create `cosmetic_manager` user	Cosmetic Lead	UAT-COS-001…006, TC-COS-*
7	Final QC Inspector	user with `cosmetic_finalqc` edit	Final QC lead	UAT-FQC-001…003
8	Spare Parts Manager	`uat_spares`	Spare Parts Manager	UAT-SPR-001…004, TC-PARTS-, *TC-RAM-*
9	Sales Executive	`uat_sales` + a `sales_manager` user	Sales lead	UAT-SAL-001…004, TC-SALE-, *TC-RET-*
10	Accounts	create custom-role user	Accounts head	(new — Section 6.10)
11	Managers / Admin	`uat_admin`	Director / Admin	UAT-RPT-001…003, UAT-ADM-001…004, UAT-RBAC-001…006
12	TRC Manager	create `trc_manager` user	TRC Manager	(new — Section 6.12)

6. Test scenarios

Format: **ID · Scenario · Expected result**. Steps for existing IDs live in `OxyPC_UAT_Scenarios_v1.0.docx`. New scenarios below are written from the current code; add them to `/qa/uat` before the cycle.

6.0 End-to-end device journey (run first, one tag, all personas)

ID	Step	Persona	Expected result (from code)
E2E-01.1	Create lot with supplier, qty, buying price, e-way bill	Inventory Manager	Lot number generated (MAX-based, collision-safe); `lots.e_way_bill` saved
E2E-01.2	GRN the lot; map GRN post-IQC	Inventory Manager	Tag counted in pipeline "GRN" (has `grn_number`)
E2E-01.3	IQC the tag, mark a part Required	IQC Inspector	Tag at `iqc` ; spec saved; dashboard IQC +1
E2E-01.4	Move to Stock In, then to Production	Inventory Manager	Stage `stock_in` → `trc_production` ; urgent flag cleared on entering Stock In
E2E-01.5	Allocate to an L1/L2 engineer	Production Manager	WorkID created; tag at `l1` ; engineer sees it in "assigned to me"
E2E-01.6	Start repair; raise part request (RAM 8GB)	Repair Engineer	`part_requests.status = requested`
E2E-01.7	Mark a second part PNA	Repair Engineer	`device_pna_parts` active row; Production PNA tile +1; L1/L2 Mark Complete blocked while PNA open

ID	Step	Persona	Expected result (from code)
E2E-01.8	Hand over the part	Spare Parts Manager	status `handed_over` ; Device Detail shows Verify button
E2E-01.9	Verify part; clear PNA; Complete to Stress	Repair Engineer	status `received` ("Part Changed"); tag at `qc_check`
E2E-01.10	Stress test **Fail** with checklist items	QC Inspector	Tag back at `l1` , assigned engineer notified; `stress_notes` filled
E2E-01.11	Fix and Complete to Stress again; Stress **Pass** (complete-to-paint)	Repair Eng., QC Inspector	Tag enters cosmetic chain
E2E-01.12	Cleaning → Putty → Dry Sanding → Masking → Painting → Water Sanding → Cosmetic Completed	Cosmetic Lead	One `StageMovement` per step; WorkIDs assigned on move
E2E-01.13	Final QC **Fail — Cosmetic**, then Move	Final QC Inspector	Tag to `final_qc_fail_hold` , then routed to Cosmetic Received by `FAIL_REASON_TO_DEPT`
E2E-01.14	Redo cosmetic; Final QC **Pass**, grade B	Final QC Inspector	Tag at `final_qc_pass_hold` ; `device_price` finalised once (base + parts + changed parts); grade saved
E2E-01.15	Move to Inventory (Ready to Sale)	Final QC Inspector / Inventory Manager	Tag at `ready_to_sale` ; dashboard Ready +1
E2E-01.16	Set min/max price; block tag for a customer	Sales Executive	Audit `TAG_PRICE_SET` ; `is_blocked = true` ; sale by other users refused
E2E-01.17	Sell from Block Tags with invoice number, warranty 6 months	Sales Executive	Sale row; tag `sold` ; company snapshot on sale; warranty expiry computed
E2E-01.18	Print invoice and delivery challan; update transport; gate-pass exit	Sales Executive	`/invoices/print/{id}` shows GST split intra/inter-state, HSN 8471; `/invoices/waybill/{id}` titled "Delivery Challan"
E2E-01.19	Record customer receipt	Accounts	Receipt at `/accounts/customer-receipts` ; sales list shows it
E2E-01.20	Process a return for credit note; approve	Sales Executive, Managers	`tag_return_status = 'Return for Credit Note'` ; Lot P&L revenue net of return
E2E-01.21	Check numbers	Managers/Admin, TRC Manager	Dashboard, Business P&L, Lot P&L, Daily Stock all reflect the tag consistently

Variants: **E2E-02** desktop and **E2E-03** TFT (forms differ — see `docs/OXYPC - Desktop-TFT - Fields.xlsx`); **E2E-04** L3/L4 path with Replacement Scrap.

6.1 IQC Inspector

ID	Scenario	Expected result
IQC-01	Register tag via OxyQC / diagnose agent	Specs auto-filled; tag at `iqc`
IQC-02	Duplicate tag or wrong lot	Live duplicate check blocks save
IQC-03	Bulk IQC upload with bad rows	Count alerts: inserted vs errors, exact tag list
IQC-04	Negative: open Sales page	Access denied / not in sidebar

6.2 Inventory Manager

ID	Scenario	Expected result
INV-01	Create, edit, trash lot	`is_trashed` set; lot gone from counts
INV-02	Stage transfer of a batch	Only `AllowedTransition` moves allowed; rejected moves listed
INV-03	Flag items urgent, then move to Stock In	Flag cleared automatically
INV-04	Change Floor / As-Is Lot sub-lot	`sub_lot_number` written to every selected tag

6.3 Production Manager

ID	Scenario	Expected result
PM-01	Summary tiles match their CSV	Same tags (shared `_prod_tile_criteria`)
PM-02	Allocate tags / buckets to engineers	WorkIDs created, notifications sent
PM-03	PNA summary CSV	Lists L1/L2 and L3/L4 tags with active PNA parts, marked by / at
PM-04	Hardware Final QC fail with no engineer	Tag parked in Production (`trc_production`)

6.4 Repair Engineer (L1/L2, L3/L4)

ID	Scenario	Expected result
REP-01	Start, pick, complete L1/L2 job	WorkID status changes; Mark Complete blocked by open PNA
REP-02	Request L3/L4 with repair note	Tag to `l3`; note shown as Repair Notes
REP-03	Multi / bulk part request	One request per tag; duplicate open request refused
REP-04	Faulty changed part	New request raised, first change kept in history
REP-05	L3/L4 Normal Scrap / Replacement Scrap	Tag scrapped or replaced; `replaced` text set

6.5 QC Inspector (Stress Test, OS install, diagnose)

ID	Scenario	Expected result
QC-01	Run stress via agent; save results	Report at `/stress/{barcode}/report`
QC-02	Fail with checklist items + note	Tag to `l1`, engineer chosen, notification; Stress Notes shown on L1/L2
QC-03	Complete to paint	Tag enters cosmetic chain
QC-04	Negative: fail to a non-L1/L2 user	Refused: "Selected user is not an active L1/L2 engineer"

6.6 Cosmetic Lead

ID	Scenario	Expected result
COS-01	Move one tag through all six stages	Each stage logs `StageMovement`
COS-02	Bulk Assign at Cosmetic Received	Selected tags advance to Cleaning and are assigned
COS-03	Bulk Assign at Painting	Pure reassignment, stage unchanged
COS-04	Fail at Cleaning / Water Sanding / Completed	Tag to `l1`, Repair Notes = " Failed — assigned to ..."

ID	Scenario	Expected result
COS-05	Skip Cosmetic → Final QC	Allowed from any cosmetic stage

6.7 Final QC Inspector

ID	Scenario	Expected result
FQC-01	Pass with grade, touch type, bucket name	Pass Hold; price finalised once
FQC-02	Fail Hardware / Software / Cosmetic, then Bulk Move	Routed to L1/L2 engineer / Stress / Cosmetic Received; unrecognised reason skipped and reported
FQC-03	Pass the same tag a second time after rework	Price **not** added again
FQC-04	Required-field validation on fail	Form refuses without reason

6.8 Spare Parts Manager

ID	Scenario	Expected result
SPM-01	Handover / Not In Stock / Procure	Status changes; Procure creates open sourcing request
SPM-02	Sourcing closed by Sales Manager then Verify	Stock credited once; request back to `requested`
SPM-03	Parts GRN (As New) with e-way bill file	Line items added; stock up
SPM-04	Add Harvest part	`source = harvest` ; Harvest count up
SPM-05	Warranty send / receive	`warranty_sent` → `warranty_replaced` , restocked

6.9 Sales Executive

ID	Scenario	Expected result
SAL-01	Ready to Sale table: auto min price	Unit price + other rate + margin (`services/ready_pricing.py`)
SAL-02	Block tags, release, sell from block	Blocked tags unsellable except from their own block
SAL-03	Multi-tag sale with price groups	One Sale row per tag, shared invoice number
SAL-04	Below-cost sale	Warning shown, not blocked
SAL-05	Mark sold via generic stage move	Not possible (`sold` excluded from `DROPDOWN_STAGES`)
SAL-06	Delivery challan and gate-pass exit (single and bulk)	Gate pass records exit
SAL-07	Returns: return for repair, paid repair, replacement, credit note	Correct `tag_return_status` ; credit note flow advances; approval required
SAL-08	Rentals	**Not found in code — mark Blocked and raise question**

6.10 Accounts

ID	Scenario	Expected result
ACC-01	Customer receipt (single and bulk upload)	Receipts saved; template download works
ACC-02	Supplier payment against PO; upload invoice	Payment complete; file stored
ACC-03	Invoice print for an inter-state sale	IGST only; intra-state shows CGST + SGST
ACC-04	Sales list export with filters	CSV matches screen
ACC-05	Daily Stock for yesterday; download saved Daily Report	Opening + movements = Closing; files present from 9 PM run

6.11 Managers / Admin

ID	Scenario	Expected result
MGR-01	Dashboard tiles vs drill-down pages	Each pipeline tile equals its `/devices?stage=…` count (except L3/L4, see 05 D5)
MGR-02	Financial Summary vs Business P&L for the same FY	Equal
MGR-03	Business P&L override for a month	FY totals and dashboard change accordingly; audit logged
MGR-04	User tickets raise / close	Ticket status Open → Closed
MGR-05	RBAC negative tests UAT-RBAC-001…006	Refused with 403 / hidden

6.12 TRC Manager

ID	Scenario	Expected result
TRC-01	Extended warranty: set and update	New expiry shown; audit logged
TRC-02	Physical audit: look up tags, submit, export stage / location / tag summaries	Record saved; CSVs match
TRC-03	Daily Stock with "Exclude Admin User"	Admin / test-user moves excluded
TRC-04	Oversight across Production, Part Master, WorkID Status	All pages open for `trc_manager` via matrix

7. Defect severity definitions

Values match `models/qa_uat.py::DefectSeverity` and `DefectPriority`.

Severity	Definition	Examples	Default priority	Fix before sign-off?
Critical	A core flow is blocked or data is wrong / lost / exposed, no workaround	Tag cannot move to Ready; sale saved with wrong price; P&L visible to engineers; 500 on a main page	P1 – Immediate	Yes

Severity	Definition	Examples	Default priority	Fix before sign-off?
High	A core flow works only with a hard workaround, or a number shown to managers is wrong	Fail reason lost; dashboard tile ≠ report; bulk action silently skips tags	P2 – High	Yes
Medium	A secondary feature fails or a core one has an easy workaround	Export column wrong; filter ignored; wrong label	P3 – Medium	No, if dated and owned
Low	Cosmetic or wording issue; no effect on data	Alignment, colour not `#0F7A37`, typo	P4 – Low	No

Status flow in the QA Dashboard: `New → Assigned → In Progress → Fixed → Retest → Closed` (or `Reopened`, `Won't Fix`).

8. Regression checklist (every release, before and after UAT fixes)

#	Check	How	Pass rule
1	Unit + workflow tests	`pytest` (`pytest.ini`, `tests/`)	All green
2	Template parse	`regression-sweep` Step 1 (now 242 templates)	0 errors
3	Route scan	`regression-sweep` Step 2	Exactly the 9 known flagged routes
4	Schema check	`python db_validator.py`	No drift
5	Auto smoke	`run_uat_excel.py` on staging	No new failures
6	Stage rules	Try one disallowed move per stage group	Refused by `validate_transition`
7	Sell guard	Try to sell a blocked tag and a non-ready tag	Refused by `validate_sale_allowed`
8	Dashboard reconcile	MGR-01, MGR-02	Equal
9	Daily Stock	ACC-05	Reconciles
10	RBAC	UAT-RBAC-001...006 + one custom role (e.g. `trc_manager`)	Correct allow / deny
11	CSRF	POST without token	403 (re-check Showstopper SEC-01)
12	Global Table	Pagination, search, sticky header on Devices, Ready to Sale, Sales	Matches Design System §07
13	Exports	Devices, Sales, Lot P&L, Daily Stock CSV / XLSX	Columns align with headers
14	Audit log	Price set, sale, return, part handover produce rows at `/admin/audit-log`	Rows present

9. Sign-off form (copy one per persona per release)

```
OxyPC Inventory Lifecycle — UAT Sign-off
-----------------------------------------------------------------
Release / version (QA Dashboard):  v_____   Build commit: ________
Environment: Staging   URL: _____________________   Dates: __/__/____ to __/__/____
Persona: _____________________   App role / login used: _______________

Scenarios in scope (IDs): ____________________________________________
Executed: ____   Passed: ____   Failed: ____   Blocked: ____

Open defects (ID / severity / agreed fix date / owner):
  1. ____________________________________________________________
  2. ____________________________________________________________

Conditions or notes: ____________________________________________________

Decision:  [ ] Accept   [ ] Accept with conditions   [ ] Reject

Business owner: _____________________  Signature: ___________  Date: __/__/____
QA lead:        _____________________  Signature: ___________  Date: __/__/____
App Dev lead:   _____________________  Signature: ___________  Date: __/__/____
-----------------------------------------------------------------
Record in QA Dashboard: /qa/uat → scenario → sign_off_by / sign_off_at
```

Release approval (once all persona forms are in):

Persona	Decision	Signed by	Date
IQC Inspector			
Inventory Manager			
Production Manager			
Repair Engineer			
QC Inspector			
Cosmetic Lead			
Final QC Inspector			
Spare Parts Manager			
Sales Executive			
Accounts			
Managers / Admin			
TRC Manager			
Release approved for production (Managers/Admin)			

10. Cycle timeline (per release)

Day	Activity
D-2	Deploy to staging; entry checks E1–E8
D1	E2E-01 walkthrough with all personas together
D2–D4	Persona scenarios; defects logged at `/qa/defects`
D5	Fixes deployed to staging; retest; regression checklist
D6	Sign-off forms; release status → `UAT Done` → `Approved`
D7	Production deploy (`commit-deploy-gate` skill); smoke on production

11. Open questions

#	Question	Owner
1	Who hosts staging and who restores the masked copy each cycle?	Admin
2	Where is `_scan_routes.py` ? Referenced by the `regression-sweep` skill but not in this checkout.	App Development Team
3	Rentals (SAL-08): in scope or not?	Sales lead

PART B

Runbook & Disaster Recovery

This is the operator's guide for the production OxyPC Inventory Lifecycle server. It covers daily checks, how to start and stop the app, common incidents, backup and restore, and disaster recovery. It builds on `docs/backup-runbook.md`, `deploy/DEPLOY.md` and `docs/superpowers/specs/2026-10-01-docker-containerization-design.md`.

Production setup at a glance (from 06 Deployment): Linux Internal Server `10.199.206.109`, app folder `/opt/oxypc`, Docker container `oxypc-app` on port 8000, nginx on port 80, PostgreSQL 17 native on the same box, database `oxypc_db`. Commands below assume an SSH session on that server in `/opt/oxypc`.

Rule: any restart or deploy follows `.claude/skills/commit-deploy-gate` — get an explicit go-ahead for this specific action first.

1. Daily checks (5 minutes, start of shift)

#	Check	How	Healthy result	Source
1	App and DB up	`curl -s http://127.0.0.1:8000/health` (or open `http://10.199.206.109/health`)	`{"status":"ok","db":"ok",…}`, HTTP 200	`routers/health.py`
2	Container healthy	`docker compose ps`	`oxypc-app` … `(healthy)`	`docker-compose.yml` healthcheck
3	Last night's backup ran	`journalctl -u oxypc-backup --since yesterday` or Admin → Database Backup panel (`/admin/backup-status`)	A file `oxypc_YYYYMMDD_23xxxx.sql.gz` under 26 h old	`deploy/oxypc-backup.service`, `routers/admin.py` l.613
4	Backup timer armed	`systemctl list-timers oxypc-backup.timer`	Next run tonight 23:00 IST	`deploy/oxypc-backup.timer`
5	Auto-deploy healthy	`journalctl -u oxypc-autodeploy --since today`	Silence, or `DEPLOYED OK` lines. No `REFUSING TO DEPLOY` / `DEPLOY UNHEALTHY`	`deploy/oxypc-autodeploy.sh`
6	Disk space	`df -h /opt /var/lib/docker /var/lib/postgresql`	Under 80 % used (proposed threshold)	docker spec §9 (44 % on 2026-10-01)
7	New errors	`docker compose logs --since 24h oxypc-app | grep -E "500 |Traceback"`	None, or known	`main.py` `generic_exception_handler`
8	Daily Stock report made	Reports page / daily report list	Yesterday's 9 PM file exists	`services/daily_stock_reports.py`

Note: `oxypc-backup.service` is `Type=oneshot`, so `systemctl is-active` showing "inactive" after a good run is **normal** (`deploy/oxypc-backup.service` header).

2. Start, stop, restart

2.1 Production (Docker)

Action	Command	Notes
Status	`docker compose ps`	
Live logs	`docker compose logs -f oxypc-app`	Console output includes startup steps and 500 tracebacks
Restart app	`docker compose restart oxypc-app`	~10 s; worst cold start seen ~9 s (`deploy/oxypc-autodeploy.sh`)
Stop app	`docker compose stop oxypc-app`	Users see nginx 502
Start app	`docker compose up -d`	Runs `db_validator` and startup backfills
Rebuild + start (deploy current checkout)	`docker compose up -d --build`	What auto-deploy does
Pause auto-deploy	`sudo systemctl stop oxypc-autodeploy.timer`	Do this before any manual change or cutover
Resume auto-deploy	`sudo systemctl start oxypc-autodeploy.timer`	
Deploy now	`sudo systemctl start oxypc-autodeploy.service`	
nginx	`sudo nginx -t && sudo systemctl reload nginx`	Config: `deploy/nginx-oxypc-internal.conf`
PostgreSQL	`sudo systemctl status postgresql` / `restart postgresql`	App reconnects (`pool_pre_ping=True` , `database.py`)

2.2 Roll back a bad deploy

1. `sudo systemctl stop oxypc-autodeploy.timer` (stop it redeploying).
2. Read the last good SHA: `cat /var/lib/oxypc-last-good-sha` .
3. `cd /opt/oxypc && git reset --hard <sha> && docker compose up -d --build` (command printed by `deploy/oxypc-autodeploy.sh` on failure).
4. Check `/health` . Then fix forward on `main` , because the timer will redeploy `main` when resumed.

Emergency fallback to bare metal (docker spec §6): `docker compose down` then `sudo systemctl enable --now oxypc.service` . Only if the host venv is intact (see Open questions).

2.3 Legacy Windows server (if ever used again)

Action	Command	Source
Start	`start_oxypc.bat` or scheduled task `OxyPC_ERP` / `OxyPC_ERP_Watchdog`	`start_oxypc.bat` , `setup_autostart.ps1` , `setup_watchdog.ps1`
Service	`nssm start|stop|status OxyPC_ERP` (or `OxyPCInventory`)	`install_service.bat` , `scripts/install_service.ps1`
Stop	`oxypc_stop.bat` (kills port 8000 listener). Avoid `stop_server.bat` : it kills **every** `python.exe`	files

3. Common incidents

Each row: **Symptom → Check → Fix**. Commands are for production unless noted.

3.1 Server down (browser cannot load the app)

Symptom	Check	Fix
Browser "can't reach site"	`ping 10.199.206.109`; `sudo systemctl status nginx`	Network/power issue → escalate to IT. nginx stopped → `sudo systemctl start nginx`
nginx "502 Bad Gateway"	`docker compose ps`; `curl -s 127.0.0.1:8000/health`	Container stopped → `docker compose up -d`. Restarting in a loop → read `docker compose logs --tail=100 oxypc-app` (often a schema error, see 3.5)
Page redirects to `https://` and fails	Was `X-Forwarded-Proto` added to the nginx vhost?	Remove it. The LAN vhost must not set it (`deploy/nginx-oxypc-internal.conf`, `main.py` `_force_https`)
New code not live after push	`journalctl -u oxypc-autodeploy -n 50`	`REFUSING TO DEPLOY: working tree has uncommitted changes` → inspect `git status` on server, commit or discard deliberately. `fetch failed` → network or deploy key
Users see old page after deploy	Ctrl+F5	HTML is `no-store`; static files are cache-busted by `ASSET_VERSION` (`main.py`, `templates_config.py`). If a JS file changed but is not in `_VERSIONED_ASSETS`, add it

3.2 Database connection errors

Symptom	Check	Fix
`/health` returns 503 `"db":"error: …"`	`sudo systemctl status postgresql`; `sudo -u postgres psql -c "select 1"`	Start PostgreSQL. Then `docker compose restart oxypc-app` if errors persist
"too many connections"	`sudo -u postgres psql -c "select count(*), state from pg_stat_activity group by state"`	Pool per process is 8 + 4 (`database.py`). Look for idle-in-transaction sessions or extra processes (old `oxypc.service`, scripts). Terminate stuck sessions; restart app
Connection refused from container	`.env` `OXYPC_DATABASE_URL` host; `pg_hba.conf`	Container uses host network and must use `localhost` / `127.0.0.1`. PostgreSQL only trusts `127.0.0.1` (docker spec §9)
Password failure after rotation	`.env` vs DB role	Update `.env`, `docker compose up -d` (env is read at container start)

3.3 403 "Access Denied" for a user

Symptom	Check	Fix
User sees 403 on a page their role should open	Master Data → Module Permissions for that role (the 403 page says this, `main.py` `_ERROR_INFO`)	Tick the module for the role. Changes reach every cache within 60 s (`main.py` cache refresh)
Custom role (e.g. `trc_manager`) can open the page but buttons are missing	Template uses `current_user.role.value == …` instead of `role_allowed()`	Code fix per `.claude/skills/custom-role-permissions`

Symptom	Check	Fix
Built-in role blocked on an admin-only page	`require_roles(UserRole.admin)` on the route	Grant the matching module in the matrix; `_matrix_grants_path` honours explicit grants (`auth/dependencies.py`)
User bounced to login on form submit	CSRF cookie missing/stale = session expired	Log in again. Sessions last 1 day (`config.py`)
User cannot log in at all	Lockout: 5 failures in 15 min (`routers/auth.py`); or user `status` inactive	Wait 15 min or have admin check User Management
"429 Too Many Requests"	Login is limited to 20/min per IP	Wait one minute

3.4 Slow pages

Symptom	Check	Fix
Whole app slow for everyone at once	Is a manual backup running? (`/admin/backup-now` blocks the single worker up to 300 s, `routers/admin.py`)	Wait for it to finish; avoid manual backups in working hours
	`docker stats oxypc-app` ; `top` for postgres	High CPU in postgres → find slow query: `select pid, now()-query_start, query from pg_stat_activity where state='active' order by 2 desc`
One list page slow	Is it a server-side DataTables page? (`serverSide: true` in template)	Check indexes exist: `\di` in psql vs `alembic/versions/20260925_1200_add_dashboard_perf_indexes.py` and `migrate_perf_indexes.py` . Re-run missing index scripts
Large export slow or cut off	Reports are capped at 5,000 rows (`routers/reports.py`)	Narrow the date range

3.5 Failed schema change / migration

Symptom	Check	Fix
Container restarts in a loop after deploy	`docker compose logs oxypc-app | grep -A20 Schema`	`db_validator` raised `RuntimeError` for an unfixable mismatch (it can add tables/columns, not change types or rename). Roll back (2.2), then fix with an Alembic revision or a `migrate_*.py` script
500 on a page with `UndefinedColumn`	Was `OXYPC_AUTO_FIX` set to 0?	Run `docker compose exec oxypc-app python db_validator.py` to diagnose; restart with auto-fix on
`alembic upgrade head` fails	`alembic current`	Production is often not stamped at head (see `20260925_1200_add_dashboard_perf_indexes.py` note). Check the revision's SQL by hand; most revisions use existence checks
Before any manual migration	—	Take a backup first (`migrate_perf_indexes.py` header: "Backup taken first")

3.6 Disk full

Symptom	Check	Fix
Uploads fail, backup fails, Postgres errors "No space left on device"	`df -h` ; `du -sh /opt/oxypc/backups /opt/oxypc/uploads /var/lib/docker`	1) `docker builder prune -f` (reclaimed 4.48 GB on 2026-10-01, docker spec §9). 2) Old images: `docker image ls` , remove unused. 3) Backups are never pruned by default: copy old ones off-server, then delete via Admin panel or set `OXYPC_BACKUP_PRUNE=1` for one run. 4) Container logs: `docker inspect --format='{{.LogPath}}' oxypc-app` and truncate if huge
Upload "Permission denied" (not disk)	`ls -ln /opt/oxypc/uploads`	Must be owned by uid 1000: `sudo chown -R 1000:1000 uploads backups static/stress_reports` (docker spec §9)

4. Backup

4.1 What is backed up today

Item	Backed up?	How	Where	Source
Database `oxypc_db`	Yes, nightly 23:00 IST (+ up to 2 min jitter; missed runs fire on boot)	`oxypc-backup.timer` → `oxypc-backup.sh` → `scripts/backup_db.py run_backup()` (`pg_dump --format=plain` , gzip)	`/opt/oxypc/backups/oxypc_YYYYMMDD_HHMMSS.sql.gz`	`deploy/oxypc-backup.*`
Database (on demand)	Yes	Admin → "Run Backup Now" (`POST /admin/backup-now` , never prunes)	same folder	`routers/admin.py` l.711, commit `6b6bf96`
`uploads/` (invoices, payment photos, documents)	**No**	—	—	no job found
`.env` and server config	**No**	—	—	no job found
FieldOps database	**No** job in repo	—	—	`fieldops_db.py`
Off-server copy	**No** (recommended only)	—	—	`docs/backup-runbook.md` , `deploy/DEPLOY.md` §7

Retention: everything is kept unless `OXYPC_BACKUP_PRUNE=1` ; then files older than 30 days are deleted (`deploy/oxypc-backup.sh` , `scripts/backup_db.py`).

4.2 Recommended additions *(proposed, not yet built)*

1. Nightly copy of the newest dump **and** a tar of `uploads/` to a second machine or cloud (`rclone` example in `docs/backup-runbook.md`).
2. Weekly backup of `/opt/oxypc/.env` to a secure password store (not git).
3. A check that alerts if the newest backup is older than 26 h.

5. Restore

Always test in a separate database first (`docs/backup-runbook.md`). The dump is plain SQL **without** `--clean` , so it must be loaded into an **empty** database. Restoring on top of the live DB will fail or duplicate data, even though `scripts/restore.sh` says it drops tables.

5.1 Test restore (monthly, from `docs/backup-runbook.md` , adapted for production)

```
cd /opt/oxypc
sudo -u postgres createdb oxypc_test
gunzip -c backups/oxypc_YYYYMMDD_HHMMSS.sql.gz | sudo -u postgres psql -d oxypc_test
sudo -u postgres psql -d oxypc_test -c "SELECT COUNT(*) FROM devices;" -c "SELECT COUNT(*)
FROM sales;"
sudo -u postgres dropdb oxypc_test
```

Record the date, file used, row counts and time taken.

5.2 Production restore

1. Get go-ahead (commit-deploy-gate). Announce downtime to users.
2. `sudo systemctl stop oxypc-autodeploy.timer`
3. `docker compose stop oxypc-app`
4. Take a safety dump of the current DB: `sudo -u postgres pg_dump -Fc oxypc_db > /opt/oxypc/backups/pre_restore_$(date +%Y%m%d_%H%M).dump`
5. Rename the current DB instead of dropping it: `sudo -u postgres psql -c "ALTER DATABASE oxypc_db RENAME TO oxypc_db_old_$(date +%Y%m%d);"`
6. Create an empty DB owned by the app role: `sudo -u postgres createdb -O <app_db_user> oxypc_db` (role name is in `OXYPC_DATABASE_URL`).
7. Load the dump: `gunzip -c backups/<file>.sql.gz | sudo -u postgres psql -d oxypc_db`
8. `docker compose up -d` — startup runs `db_validator` and adds any tables/columns newer than the dump.
9. Optional: `docker compose exec oxypc-app alembic upgrade head` (per `docs/backup-runbook.md`).
10. Check `/health` , log in, open Devices and Sales lists, compare counts with step 7 of the test restore.
11. `sudo systemctl start oxypc-autodeploy.timer`
12. Keep `oxypc_db_old_*` until the business confirms, then drop it.

Uploaded files are not in the dump. If `uploads/` was lost, restore it separately from whatever copy exists (none is automated today).

6. Disaster recovery

6.1 Scenarios

Scenario	Impact	Recovery today	Gap
App crash / bad deploy	App down	Docker auto-restart; rollback per 2.2	No auto-rollback (deliberate)

Scenario	Impact	Recovery today	Gap
PostgreSQL corruption	Data loss since last dump	Restore last nightly dump (section 5)	Up to ~24 h of data lost
Server disk or hardware failure	App, DB **and backups** lost together	Rebuild a Linux box per `deploy/DEPLOY.md` + Docker files; restore DB from an off-server copy	**No off-server copy exists**, so full loss is possible
Ransomware / accidental mass delete	Data damaged	Restore an older dump (all are kept by default)	Backups sit on the same server and could be hit too
Office network down	Users cannot reach the LAN app	None	No remote access path documented

6.2 Rebuild on new hardware (outline)

1. Install Ubuntu, PostgreSQL 17, nginx, Docker Engine + Compose (`deploy/DEPLOY.md` §2-3; versions from docker spec §9 and `Dockerfile`).
2. Create DB and role; restore the latest dump (section 5.2 steps 6-7).
3. Clone the repo to `/opt/oxypc` with the read-only deploy key; restore `.env` .
4. `mkdir uploads backups static/stress_reports && chown -R 1000:1000 …` ; restore `uploads/` if a copy exists.
5. `docker compose up -d --build` ; install nginx vhost `deploy/nginx-oxypc-internal.conf` .
6. Install timers: `oxypc-backup.*` , `oxypc-autodeploy.*` (headers of each file list the commands).
7. Point users at the new IP (or move `10.199.206.109` to the new box).

6.3 RPO and RTO

Measure	Defined?	What the setup implies today
RPO (max data loss)	**No**	Up to 24 h (one nightly dump); total loss if the server disk dies
RTO (max downtime)	**No**	Minutes for app faults; unknown (likely a day or more) for hardware loss, as no standby or spare exists

These must be agreed with the business. See Open questions.

---

7. Contacts

Role	Name	Phone / email	When to call
Business owner / Director	Pankaj Sehgal (`CLAUDE.md`)	*TBD*	Approval for restarts, restores, data fixes
App development team	*TBD*	*TBD*	Code faults, deploys, rollbacks
Server / IT admin (Internal Server)	*TBD*	*TBD*	Hardware, network, OS, SSH, disk
Database admin	*TBD*	*TBD*	Restores, performance
Internet / network vendor	*TBD*	*TBD*	LAN or WAN outage

Role	Name	Phone / email	When to call
Escalation if nobody answers	*TBD*	*TBD*	

Open questions

1. What RPO and RTO does the business accept? Is a 24-hour data-loss window acceptable?
2. Where should the off-server backup copy go (NAS, Google Drive, S3), and who owns it?
3. Does the host still have a working `/opt/oxypc/venv` for `oxypc-backup.sh` and for the bare-metal rollback? The docker spec says nightly backups run "as root", while `deploy/oxypc-backup.service` says `User=www-data`.
4. Is there a monitoring tool watching `/health` from outside the server (the docstring mentions Uptime Robot/Grafana)?
5. Who has SSH access to `10.199.206.109` and sudo rights? Who holds the GitHub deploy key?
6. Is the FieldOps database backed up, and by whom?
7. Has a full test restore ever been done? When, and how long did it take?
8. What is the plan if the office LAN or the server room loses power for a day?

PART C

Deployment & Environments

This document explains where the app runs, how it is started, how configuration and secrets are supplied, how the database schema changes, and how a change reaches production. It cites the file behind each statement. Secret **values** are never shown; only variable names.

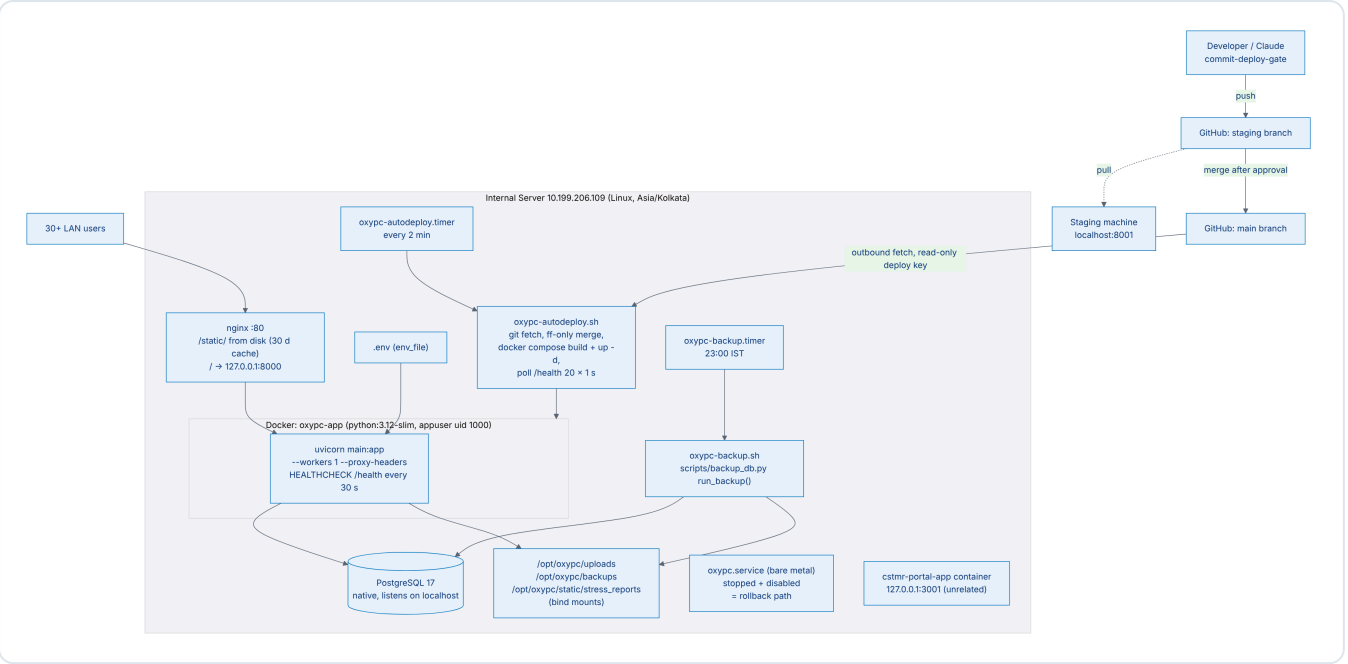
Key point. Production is no longer a Windows laptop. Since 1 Oct 2026 it is a Linux "Internal Server" (`10.199.206.109`) running the app in a Docker container behind nginx, with PostgreSQL installed natively on the same box (`docs/superpowers/specs/2026-10-01-docker-containerization-design.md` §9). The Windows scripts in the repo root describe the earlier laptop setup and the standalone UAT installer. They are kept here as "legacy / local" paths.

1. Environments

Environment	URL	Host and runtime	Code source	Database	Source
Production	`http://10.199.206.109` (plain HTTP, LAN only)	Linux Internal Server, `/opt/oxypc` ; nginx :80 → Docker container `oxypc-app` (uvicorn, 1 worker, host network, port 8000)	GitHub `main` (`Bluemonk123/oxypc-inventory`), pulled every 2 min	PostgreSQL 17, native on the same host	`deploy/nginx-oxypc-internal.conf` , `docker-compose.yml` , `Dockerfile` , `deploy/oxypc-autodeploy.sh` , design system release table
Staging	`http://localhost:8001`	Staging machines run `RUN_STAGING.bat` (Windows) or `START_STAGING.command` (Mac)	`staging` branch	Not stated	`qa_docs/design_system/design_system.html` (Environments row and "From change to production" step 3)
Local / developer	`http://localhost:8000`	`python main.py` or `uvicorn main:app`	Any branch	Local PostgreSQL (`oxypc_db`)	`README.md` Quick Start, `.claude/launch.json`
Legacy Windows server	`http://192.168.1.100:8000`	Windows laptop, uvicorn via scheduled task / NSSM service / watchdog	Local folder `C:\Users\Pankaj.sehgal\Claude\Oxypc\oxypc-inventory`	Local PostgreSQL	`CLAUDE.md` , `open_firewall.ps1` , `run_server.ps1`
Standalone UAT installer	`http://localhost:8000` (+ optional Cloudflare tunnel)	`OxyPC_UAT_Setup.exe` bundling Python 3.11.9 + PostgreSQL 16.6 (port 5433) + cloudflared	Build machine	Bundled PostgreSQL	`oxypc_setup.iss` (version `1.0-UAT`), `prepare_bundle.bat` , `BUILD_ME.bat`
Public VPS (planned)	`https://erp.oxypc.in` (placeholder)	Ubuntu VPS, nginx + certbot, systemd	—	—	`deploy/DEPLOY.md` ("Replace erp.oxypc.in with your real subdomain")

`RUN_STAGING.bat` and `START_STAGING.command` are **not** in the repository. See Open questions.

2. Production deployment diagram



Sources: `deploy/oxypc-autodeploy.{timer,service,sh}`, `deploy/oxypc-backup.{timer,service,sh}`, `deploy/nginx-oxypc-internal.conf`, `docker-compose.yml`, `Dockerfile`, docker spec §9 (other container, rollback path, PostgreSQL 17 noted in `Dockerfile`).

3. How the server runs

3.1 Production (Linux + Docker)

Piece	What it does	Source
`Dockerfile`	Two-stage build on `python:3.12-slim`; installs `postgresql-client-17` so the in-app "Run Backup Now" button has `pg_dump`; runs as non-root `appuser` (uid 1000); `HEALTHCHECK` calls `/health`; `CMD uvicorn main:app --host 0.0.0.0 --port 8000 --workers 1 --proxy-headers`	`Dockerfile`, commit `6b6bf96`
`docker-compose.yml`	One service `oxypc-app`, `restart: unless-stopped`, `network_mode: host`, `env_file: .env`, bind mounts for `uploads/`, `backups/`, `static/stress_reports/`	`docker-compose.yml`
`.dockerignore`	Keeps local cruft (venvs, `wa-service/`, agent build output, Office files) out of the image	`.dockerignore`, docker spec §9
nginx	LAN vhost, no TLS; deliberately does **not** set `X-Forwarded-Proto` (otherwise the app's HTTPS redirect would break plain-HTTP access); serves `/static/` from disk with gzip and 30-day cache; 50 MB upload limit; 120 s proxy timeout	`deploy/nginx-oxypc-internal.conf`
Why one worker	In-memory caches (`_PERM_CACHE`, transitions) and the in-memory rate limiter are per process	`Dockerfile` comment, `main.py` `__main__` comment

Piece	What it does	Source
Rollback path	`oxypc.service` (bare-metal venv) kept installed but disabled	docker spec §6 and §9

Conflict to resolve: `deploy/oxypc.service` and `services/oxypc-inventory.service` say 4 workers are safe because of the 60-second cache refresh. `Dockerfile` says one worker "is a hard requirement". Production uses one worker (Docker). See Open questions.

3.2 Windows scripts (legacy server, local and UAT)

All Windows scripts hard-code the path `C:\Users\Pankaj.sehgal\Claude\Oxypc\oxypc-inventory` and port 8000.

Script	Purpose	Notes	Source
`start_server.bat`	Endless loop: start uvicorn on `127.0.0.1:8000`, restart 5 s after exit; logs to `server_service.log`	Binds to localhost only, so not reachable from LAN	file
`start_oxypc.bat`	One-shot uvicorn on `0.0.0.0:8000`, 1 worker, logs to `server.log`	Used by the watchdog	file
`run_server.ps1`	Supervisor loop around uvicorn (`C:\Python314\python.exe`), restart after 3 s	Launched by the `OxyPC_ERP` scheduled task	file
`setup_autostart.ps1`	Registers scheduled task `OxyPC_ERP` at logon, restart up to 999 times	Run elevated once	file
`watchdog_server.ps1`	Pings `http://127.0.0.1:8000/auth/login`; if down, kills the port-8000 listener and starts `start_oxypc.bat`	Logs to `watchdog.log`	file
`setup_watchdog.ps1`	Registers `OxyPC_ERP_Watchdog` task: at logon and every 30 min; **removes** the `OxyPC_ERP` task	So watchdog and supervisor are alternatives, not both	file
`install_service.bat`	NSSM service `OxyPC_ERP` running `C:\Python313\python.exe main.py`; logs rotated at 10 MB	Needs `nssm.exe` on PATH	file
`scripts/install_service.ps1`	Downloads NSSM 2.24, installs service `OxyPCInventory` wrapping `scripts/start_server.bat` (venv, `0.0.0.0:8000`)	Second, different service name	file
`setup_server_ip.bat`, `open_firewall.ps1`	Open firewall port 8000; force static IP `192.168.1.100`, DNS `8.8.8.8`	Changes network settings	files
`stop_server.bat`, `oxypc_stop.bat`	Kill uvicorn / ngrok / cloudflared; `stop_server.bat` kills **every** `python.exe`	Use with care	files
`launcher.py`, `oxypc_launcher.bat`	Runs the app plus a Cloudflare tunnel; expects PostgreSQL already running		`launcher.py` header
`build_all.bat`, `BUILD_ME.bat`, `prepare_bundle.bat`, `build_installer.bat`, `oxypc_setup.iss`, `oxypc_setup_light.iss`	Build `dist\OxyPC_UAT_Setup.exe` with Inno Setup 6; bundle Python 3.11.9, PostgreSQL 16.6-1 (port 5433), cloudflared	Version `1.0-UAT`; Windows 10 1809+ (`MinVersion=10.0.17763`)	files

Script	Purpose	Notes	Source
`build.spec`	PyInstaller spec → `dist/OxyPC_Inventory.exe`		`README.md` "Build EXE"
`installer.py`	Browser-based Linux installer on port 8080; clones `TheCodeOrbit/oxypc` into `/opt/oxypc`	Points at a different GitHub repo	`installer.py` header

3.3 Linux bare metal (pre-Docker, kept for rollback)

`deploy/oxypc.service` runs `/opt/oxypc/venv/bin/uvicorn main:app --host 127.0.0.1 --port 8000 --workers 4` as `www-data`, with `Restart=always`. `deploy/DEPLOY.md` is the full VPS runbook (PostgreSQL, venv, systemd, nginx + certbot, hardening list).

---

4. Configuration and secrets

4.1 How settings are resolved

`config.py` loads `.env` (python-dotenv), then reads each setting in this order: **environment variable → `config.ini` → hard-coded default**. On first run `main.py` writes a `config.ini` if none exists (`write_default_config`), and that file contains the secret key and API key in plain text.

`.env` and `config.ini` must never be committed (`config.py` comment). Templates: `.env.example`, `deploy/.env.example`.

4.2 Environment variables (names only)

Variable	Used for	Read in
`OXYPC_DATABASE_URL`	Main DB URL (`postgresql+asyncpg://…`)	`config.py`, `scripts/backup_db.py`
`OXYPC_SECRET_KEY`	JWT signing key	`config.py`
`OXYPC_API_KEY`	OxyQC machine API key	`config.py`
`OXYPC_TOKEN_EXPIRE_MINUTES`	**No longer read.** Session length is fixed at 1440 min in code (commit `d8bc498`)	`.env.example` only
`OXYPC_REFRESH_DAYS`	Refresh token days (default 7)	`config.py`
`OXYPC_PORT`, `OXYPC_HOST`	Bind port/host for `python main.py`	`config.py`
`OXYPC_COOKIE_SECURE`	Secure flag on auth cookies (needs HTTPS)	`config.py`
`OXYPC_ALLOWED_ORIGINS`	CORS origins for `/api/v1/*`	`config.py`
`OXYPC_DB_POOL_SIZE`, `OXYPC_DB_MAX_OVERFLOW`	Pool size (default 8 + 4)	`database.py`
`OXYPC_AUTO_FIX`	Let `db_validator` fix schema at startup (default on)	`main.py`

Variable	Used for	Read in
`OXYPC_DEBUG`	Show exception detail on error pages (LAN dev only)	`main.py`
`OXYPC_OPEN_BROWSER`	Open a browser when run via `python main.py`	`main.py`
`OXYPC_FIELDOPS_DATABASE_URL` / `FIELDOPS_DATABASE_URL`	FieldOps DB; unset → `/fieldops` disabled	`fieldops_db.py`
`FIELDOPS_ADMIN_USERNAME`, `FIELDOPS_ADMIN_PASSWORD`, `FIELDOPS_DEMO_PASSWORD`, `FIELDOPS_DB_POOL_SIZE`, `FIELDOPS_DB_MAX_OVERFLOW`	FieldOps accounts and pool	`fieldops_db.py`
`OXYPC_CARE_API_BASE`	Care Agent → server base URL	`care_agent_windows/`
`OXYPC_BACKUP_PRUNE`	`1` lets the nightly job prune old backups (default off)	`deploy/oxypc-backup.sh`
`OXYPC_DB_HOST`, `OXYPC_DB_PORT`, `OXYPC_DB_NAME`, `OXYPC_DB_USER`, `OXYPC_BACKUP_DB_PASS`	Root-level `backup_db.py` (older script)	`backup_db.py`
`PGPASSWORD`, `OXYPC_BACKUP_DIR`, `OXYPC_RETENTION_DAYS`	Shell backup/restore scripts	`scripts/backup.sh`, `scripts/restore.sh`

Risky defaults in code (values not repeated here): `config.py` has a fallback DB URL with a password, and a fallback OxyQC API key marked "change-me". If `OXYPC_SECRET_KEY` is unset and no `config.ini` exists, a random key is generated and written to `config.ini`. `deploy/DEPLOY.md` "MUST-HARDEN" lists rotating all of these.

5. Database schema changes

Three mechanisms exist side by side. Production relies mainly on the first.

Mechanism	What it does	When it runs	Source
`db_validator.validate_and_fix()`	Compares ORM models to the live DB. Creates missing tables, `ADD COLUMN IF NOT EXISTS` for missing columns, reseeds `stage_master` and `allowed_transitions`. Raises (and aborts startup) if something cannot be fixed	Every app start (so every deploy) when `OXYPC_AUTO_FIX=1` (default). Also `python db_validator.py` to diagnose only	`db_validator.py` header, `main.py` startup, `deploy/oxypc-autodeploy.sh` comment
Alembic	36 revisions from `20260331_1600_ba22dd09a6a3_initial_baseline` to `20261013_1200_dealers_sub_category_website_cid`. `alembic/env.py` reads the URL from `config.py`	Manual `alembic upgrade head`. Not run by the Docker auto-deploy	`alembic/versions/`, `alembic.ini`
One-off scripts	51 `migrate_*.py` (e.g. `migrate_perf_indexes.py` creates/drops indexes), 8 `backfill_*.py`, `upgrade_db.py` (adds enum values and columns, `create_all`, stamps Alembic), `setup_db.py` (first install)	Manual, one at a time	repo root

Mechanism	What it does	When it runs	Source
Startup backfills	Idempotent one-time data fixes (GRN-Lot links, tag backfills, size units, Product Master)	Every app start	`main.py` startup

Known consequences:

- `db_validator` only **adds**. It does not rename, drop, change types, or add indexes. Those still need Alembic or a `migrate_*.py` script.
- Alembic is not reliably applied in production. `20260925_1200_add_dashboard_perf_indexes.py` says indexes "declared in an earlier migration (f991834630d1) … [were] never actually present on the live schema".
- `CLAUDE.md` rule: no schema change without dbdiagram.io approval (`docs/schema.dbml` ; design system step 1).

---

6. Release process

6.1 Branches

Branch	Role	Evidence
`staging`	Work lands here first; staging machines load it at `localhost:8001`	Design system step 3; `origin/staging` = `55ce58a`
`main`	Production. Auto-deploy polls `origin/main`	`deploy/oxypc-autodeploy.sh` ; `origin/main` = `bfca61d` (behind staging at time of writing)

6.2 Steps (from `qa_docs/design_system/design_system.html` "From change to production")

#	Step	Gate	Source
1	Database approval	New table/column drafted as `.dbml` , approved in dbdiagram.io	design system, `CLAUDE.md`
2	Build and verify	`verify-feature` skill (compile, Jinja parse, `db_validator.py` , httpx script); `regression-sweep` if `base.html` , `app.css` , a macro or a Jinja global changed	`.claude/skills/verify-feature` , `.claude/skills/regression-sweep`
3	Staging	Push to `staging` ; test at `localhost:8001` with Ctrl+F5	design system
4	Approval	Explicit "push" for **this** batch; an earlier yes never carries over	`.claude/skills/commit-deploy-gate/SKILL.md`
5	Production	Merge to `main` . Auto-deploy picks it up within ~2 min (design system says 2-5 min)	`deploy/oxypc-autodeploy.timer` , design system
6	Docs	Update the design system HTML and rebuild the PDF in the same change	design system

6.3 What auto-deploy does

From `deploy/oxypc-autodeploy.sh` :

1. `git fetch origin main` . Exit quietly if nothing new.

2. **Refuse** to deploy if the server working tree has uncommitted changes.
3. Save the current SHA to `/var/lib/oxypc-last-good-sha`.
4. `git merge --ff-only origin/main`.
5. `docker compose build` then `docker compose up -d`.
6. Poll `http://127.0.0.1:8000/health` once a second for up to 20 s.
7. On failure, print the rollback command (`git reset --hard <last good>` + `docker compose up -d --build`). It does **not** roll back automatically, by design.

Watch it with `journalctl -u oxypc-autodeploy -f`; pause it with `systemctl stop oxypc-autodeploy.timer` (`deploy/oxypc-autodeploy.service` header).

6.4 Health checks

Check	What it tests	Source
`GET /health`	Runs `SELECT 1`. 200 `{"status":"ok","db":"ok",…}` or 503 `degraded`. Exempt from HTTPS redirect	`routers/health.py`, `main.py` `_force_https`
Docker `HEALTHCHECK`	`/health` every 30 s, 3 retries	`Dockerfile`, `docker-compose.yml`
`GET /api/v1/health`	API-layer health	`routers/api_v1/health.py`
`GET /iqc/api/health`	OxyQC connectivity	`routers/iqc_api.py` header
`GET /admin/backup-status`	Age and size of the newest backup file (admin only)	`routers/admin.py` l.613
`/admin/system/health`	**Not found in the code.** No route with this path exists on staging @ 55ce58a	grep of `routers/`, `templates/`

7. CI/CD in GitHub

Workflow	Trigger	What it does	Status note
`.github/workflows/build-agent.yml`	Push to `main` touching `oxyqc-agent/oxyqc_agent.py` or its `.spec`	Builds `Diagnose_Device_Agent.exe` on `windows-latest` with Python 3.12 and commits it	Active per file
`.github/workflows/deploy.yml`	Push to `main`	On a **self-hosted** runner: pip install into `/opt/oxypc/venv`, rsync, `alembic upgrade head`, force-run `db_validator`, `systemctl restart oxypc`, curl `/health`	Targets the bare-metal service that was disabled at the Docker cutover. May be stale

There is no CI job that runs `pytest` (no workflow calls it).

Open questions

1. Where are `RUN_STAGING.bat` and `START_STAGING.command`? Which database does staging use (a copy of production, or seed data)? Who owns the staging machines?
2. Is `.github/workflows/deploy.yml` still connected to a self-hosted runner? If yes, it would try to restart the disabled `oxypc.service` on every push to `main`.
3. Workers: should production stay at 1 worker (`Dockerfile`) or move to 4 (`deploy/oxypc.service`)? Moving needs a shared rate-limit store and a single owner for the daily report job.
4. Is Alembic `alembic_version` on production at head? What is the policy: Alembic for every change, or `db_validator` plus scripts?
5. Is there a public or Railway deployment (`README.md` names `https://app.oxypc.com/fieldops`; Railway is mentioned in `main.py`, `limiter.py`, `deploy/oxypc-autodeploy.timer`)? If so, what is its database and release flow?
6. Is `/admin/system/health` a planned page? It is not in the code.
7. Does the nightly backup still run with `/opt/oxypc/venv/bin/python` on the host after the Docker cutover? (`deploy/oxypc-backup.sh` assumes the host venv exists.)
8. Who holds the server SSH keys and the GitHub read-only deploy key, and where are `.env` values stored outside the server?

PART D

Non-Functional Requirements

This document lists the quality requirements for the OxyPC Inventory Lifecycle app. Each requirement has an ID (NFR-xx), the **current state** found in the code, a **target**, and the **gap**. Targets marked *(proposed)* are not written down anywhere in the repo; they are suggestions for the business to confirm. Unknown numbers are not invented; they are listed under Open questions.

1. Summary scorecard

ID	Area	Status
NFR-01	Large-list performance	Partly met
NFR-02	Page and dashboard response time	Unmeasured
NFR-03	Database indexing	Partly met
NFR-04	Concurrency and capacity	Partly met
NFR-05	Availability and auto-restart	Met for app; single server
NFR-06	Backup	Partly met
NFR-07	Restore and recovery	Not tested in repo
NFR-08	Authentication and sessions	Partly met
NFR-09	Authorisation	Met, with drift
NFR-10	Transport and data protection	Not met (LAN HTTP)
NFR-11	Audit trail	Met
NFR-12	Browser support	Met (desktop)
NFR-13	Display scale (90 % zoom)	Met
NFR-14	Phone and tablet use	Partly met
NFR-15	Accessibility	Not defined
NFR-16	Logging and monitoring	Partly met
NFR-17	Maintainability	Partly met
NFR-18	Testability and release safety	Partly met
NFR-19	Data integrity (soft delete, schema)	Met, with drift

Status legend: **Met** = code meets the stated target. **Partly met** = some evidence, clear gaps. **Not met / Not defined** = no evidence or no target.

2. Performance

NFR-01 · Large lists stay fast

Requirement	Lists with thousands of devices, tags or sales load and filter without freezing the browser or the server.
Current state	13 heavy list pages use server-side DataTables (`serverSide: true`), e.g. `templates/devices/list.html` , `templates/iqc/list.html` , `templates/sales/list.html` , `templates/grn/records.html` , `templates/gate_pass/list.html` . `static/js/global-table.js` adds the shared UX on top (12 rows default, sticky first column, cell clamping). Repair and QC lists got `page` / `page_size` pagination (`docs/superpowers/plans/2026-05-08-sprint21-pagination-softdelete.md`). CSV/Excel report exports are capped at `MAX_EXPORT_ROWS = 5_000` (`routers/reports.py` l.40). The LAN nginx has larger proxy buffers because "some list pages render multi-MB HTML responses" (`deploy/nginx-oxypc-internal.conf`).
Target *(proposed)*	Every list that can exceed ~1,000 rows uses server-side paging; no page sends more than ~1 MB of HTML.
Gap	229 of 242 templates do not use server-side paging; some still render full tables in HTML. No inventory of which pages exceed the size budget. The 5,000-row export cap can silently truncate (a warning exists at `routers/reports.py` l.882 for one report).

NFR-02 · Page and dashboard response time

Requirement	Common pages respond quickly for floor staff.
Current state	Dashboard counts were batched into one grouped query (PERF-3, `docs/superpowers/plans/2026-05-14-sprint22-dashboard-perf-report-limits-role-checks.md`). Sprint 18 plan aimed to cut dashboard load "from 3–8 s to under 1 s" (`docs/superpowers/plans/2026-04-27-sprint18-20-performance-smartux.md`); no measured result is recorded. Static files are served by nginx with gzip and a 30-day cache; measured on `10.199.206.109` : `bootstrap.min.css` 30.3 → 10.7 ms, `chart.umd.min.js` 40.1 → 18.4 ms, `app.js` 8.0 → 0.8 ms (`deploy/nginx-oxypc-internal.conf`). App responses are gzip-compressed above 1,000 bytes (`main.py`). A stress-test module exists (`routers/stress_api.py` , `stress_runner.py`).
Target *(proposed)*	95th percentile under 2 s for list and dashboard pages at 30 concurrent users.
Gap	No agreed target. No request-time logging or APM, so there is no baseline.

NFR-03 · Database indexing

Requirement	Hot filter, join and group-by columns are indexed.
Current state	Index migrations: `20260426_0940_f991834630d1_add_missing_fk_indexes.py` (9 indexes), `20260427_1811_b62e6ba33486_add_composite_indexes_sprint18.py` (4), `20260925_1200_add_dashboard_perf_indexes.py` (6), plus `migrate_perf_indexes.py` (adds `ix_device_location_logs_location_id` , drops a duplicate on `devices.location_id`).
Target	Every index declared in models or migrations exists in production.
Gap	The September migration states the April indexes were "never actually present on the live schema". Production index state is not checked by any tool; `db_validator.py` checks tables and columns only.

NFR-04 · Concurrency and capacity

Requirement	30+ concurrent LAN users (`CLAUDE.md`) without errors or 429s.

Current state	Production runs **one** uvicorn worker (async) in Docker (`Dockerfile`). DB pool 8 + 4 overflow per process (`database.py`). Rate limit 2,000 requests/min per session, 20/min per IP on login (`limiter.py` , `routers/auth.py` l.55); keyed per session so one NAT'd office IP is not one shared bucket. Per-process caches refreshed every 60 s (`main.py`).
Target *(proposed)*	50 concurrent users with headroom; no request blocked by another user's long task.
Gap	Any blocking call stalls everyone on a single worker. Example: `POST /admin/backup-now` runs `subprocess.run(... timeout=300)` synchronously inside an async route (`routers/admin.py` l.711-727), so the app can freeze for the whole backup. Scaling to more workers needs a shared rate-limit store and a single owner for the 9 PM report job (`services/daily_stock_reports.py`).

3. Availability and recovery

NFR-05 · Availability and auto-restart

Requirement	The app restarts itself after a crash or reboot, during working hours and after power loss.
Current state	Production: `restart: unless-stopped` and a Docker `HEALTHCHECK` on `/health` every 30 s (`docker-compose.yml` , `Dockerfile`). Auto-deploy waits up to 20 s for `/health` and reports unhealthy deploys, without auto-rollback (`deploy/oxypc-autodeploy.sh`). Rollback path: the disabled `oxypc.service` (docker spec §6). Legacy Windows: `watchdog_server.ps1` every 30 min, `setup_autostart.ps1` at logon, NSSM services (`install_service.bat` , `scripts/install_service.ps1`).
Target *(proposed)*	Available during business hours; unplanned downtime under 30 min per incident.
Gap	No uptime target or monitoring outside the box (`routers/health.py` mentions Uptime Robot but none is configured in the repo). One physical server holds app, database and backups. `restart: unless-stopped` restarts a crashed container but Docker does not restart a container that is merely "unhealthy".

NFR-06 · Backup

Requirement	Database and uploaded files are backed up daily and kept off the server.
Current state	Nightly systemd timer at 23:00 IST with `Persistent=true` (`deploy/oxypc-backup.timer`), runs `scripts/backup_db.py run_backup()` → `backups/oxypc_YYYYMMDD_HHMMSS.sql.gz` (plain SQL, gzipped). Pruning is **off** unless `OXYPC_BACKUP_PRUNE=1` ; a dump is "~2 MB" (`deploy/oxypc-backup.sh`). Admins can run, list, download and delete backups (`routers/admin.py` l.613-727, audited).
Target *(proposed)*	Daily DB **and** `uploads/` backup, with at least one copy off the server, checked daily.
Gap	Only the database is dumped; `uploads/` (invoices, payment photos, KYC, IQC images) has no backup job. Off-site copy is only "recommended" (`docs/backup-runbook.md` , `deploy/DEPLOY.md` §7). Three schedules are documented (02:00 in `docs/backup-runbook.md` and `scripts/setup_backup_task.bat` , 23:00 in `deploy/oxypc-backup.timer` and `deploy/DEPLOY.md` , 23:00 in root `backup_db.py`), and root `backup_db.py` writes `oxypc_db_YYYY-MM-DD_…` names that the admin panel's filename pattern does not list.

NFR-07 · Restore and recovery

Requirement	A tested way to restore to a known point within an agreed time.
Current state	`scripts/restore.sh` pipes a dump into `psql` after typing `RESTORE`. `docs/backup-runbook.md` describes a monthly test restore into `oxypc_test`.
Target	RPO and RTO agreed and tested (see 08 Runbook & DR).
Gap	No RPO/RTO defined. No record of a test restore. The dump is made without `--clean`, so restoring into a non-empty database fails or duplicates; `restore.sh` says it "will DROP all tables" but does not.

4. Security (summary)

Full detail lives in 04 Security & Access Design.

NFR-08 · Authentication and sessions

Current state	JWT in an `httponly`, `samesite=strict` cookie (`routers/auth.py` l.115). Session fixed at 1 day (`config.py`, commit `d8bc498`). bcrypt 5.0.0 password hashing (`auth/dependencies.py`). Lockout after 5 failed logins for 15 min (`routers/auth.py` l.37-38). Login 20/min per IP.
Target *(proposed)*	As now, plus MFA for admins (`deploy/DEPLOY.md` hardening item 9).
Gap	`secure` flag is off unless `OXYPC_COOKIE_SECURE=1`, which needs HTTPS (not on the LAN). No MFA.

NFR-09 · Authorisation

Current state	Built-in roles + admin-created custom roles + Permission Matrix (`auth/dependencies.py` `require_roles`, `_matrix_grants_path`, `require_module_perm`). Partner, care-agent and API-key auth are separate (`auth/partner_auth.py`, `auth/care_auth.py`, `auth/api_key.py`).
Gap	15 template checks still hard-code `current_user.role.value` instead of `role_allowed()` (e.g. `templates/crm/contacts/list.html` l.52), so custom roles may not see buttons their backend allows.

NFR-10 · Transport and data protection

Current state	Production is plain HTTP on the LAN (`deploy/nginx-oxypc-internal.conf`). nginx sets `X-Frame-Options`, `X-Content-Type-Options`, `Referrer-Policy`. CSRF double-submit on all writes (`verify_csrf`). PostgreSQL trusts only `127.0.0.1` (docker spec §9).
Gap	No TLS, so passwords and cookies cross the LAN in clear text. `/uploads` is mounted as public static files with no login check (`main.py` l.136); file names are UUIDs (`routers/accounts.py` l.43) but anyone with the link can open them. Code holds default credentials as fallbacks (`config.py`).

NFR-11 · Audit trail

Current state	`services/audit_engine.audit()` called 242 times in routers/services; records user, IP (from `X-Forwarded-For`), table, record, old/new values. Rule: audit **before** `db.delete()` (`CLAUDE.md`). Admin audit-log page (`routers/admin.py`).

Gap	No retention policy for audit rows. No check that every `db.delete()` (36 call sites) is preceded by an audit call.

5. Usability, devices and accessibility

NFR-12 · Browser support

Current state	`static/js/global-table.js` (point 8) states the table works "identically in Chrome, Edge, Safari, and Firefox", and fixed a cross-browser cache issue on 2026-09-03 via `ASSET_VERSION` cache busting (`templates_config.py` , `tests/test_asset_version_cache_busting.py`). All libraries are vendored (`static/vendor/`), so pages work without internet. HTML is sent `no-store` so a deploy is seen at once (`main.py` `_no_store_pages`).
Target *(proposed)*	Latest Chrome and Edge on Windows as primary; Safari and Firefox best effort.
Gap	No written support matrix or minimum versions.

NFR-13 · Display scale (90 % app zoom)

Current state	`body { zoom: 0.9 }` via `--app-zoom` (`static/css/app.css` l.12-20). Viewport-height layout uses `--full-vh = 100vh / 0.9` . Modal backdrop and Chart.js hover were fixed for zoom (`static/css/app.css` l.23-30, `static/js/chart-zoom-fix.js` , commit `eb8b4e2`).
Gap	CSS `zoom` behaves differently across engines (Firefox only gained support recently); new components must remember `--full-vh` . Effective table text is 14 px × 0.9 ≈ 12.6 px (`global-table.js` point 6), which may be small for some users.

NFR-14 · Phone and tablet use

Current state	`base.html` has a responsive viewport tag; `app.css` has one `@media (max-width: 768px)` block. Dedicated mobile PWAs exist for telecalling (`static/m/telecalling/` , with offline queue), trade partners (`static/partner/`), and Reliance FieldOps (`fieldops_app/`). Gate Pass supports scan auto-select (commit `170c51f`).
Target *(proposed)*	Security-gate staff can confirm exits on a phone; IQC and stock pages usable on a tablet.
Gap	Main staff pages are desktop-first; there is no mobile layout for Gate Pass or scanning pages and no phone test in the UAT plan. Whether gate staff actually use phones is not recorded.

NFR-15 · Accessibility

Current state	About 200 `aria-` attributes across templates (grep). Bootstrap 5 defaults. Global Table puts the full text of clamped cells in `title=` tooltips.
Target *(proposed)*	WCAG 2.1 AA for colour contrast, keyboard use and form labels on core floor pages.
Gap	No accessibility target or audit. Colour-only status pills and small effective font size are likely issues.

6. Operations and engineering quality

NFR-16 · Logging and monitoring

Current state	Unhandled 500s go to console and to `errors.log` next to `main.py` (`main.py` l.112-120). Startup prints a status line per step. In Docker, console output is visible via `docker compose logs` . Auto-deploy and backup runs are in the systemd journal.
Target *(proposed)*	Errors and slow requests retained for 30 days on persistent storage; alert when `/health` fails or no backup in 26 h.
Gap	In the container `errors.log` is written to `/app/errors.log` , which is **not** a bind mount, so it is lost on every rebuild (every deploy). No log rotation is configured for `errors.log` or Docker's log driver (`docker-compose.yml`). No access log with timings, no alerting.

NFR-17 · Maintainability

Current state	Clear layering (routers / models / services / templates), project skills that encode rules (`.claude/skills/`), and a design system document (`qa_docs/design_system/design_system.html`).
Gap	`main.py` is 868 lines with many one-time backfills in startup. The repo root holds 51 `migrate_*.py` , 8 `backfill_*.py` , 12 `verify_*.py` , 9 `seed_*.py` , 7 `_*.py` scratch scripts, Office documents and logs (`server_log.txt` , `test_results.txt`). Three ways to change the schema (see 06 §5). Two Windows service names (`OxyPC_ERP` , `OxyPCInventory`). Stray `routers/repair.py.bak` . Dead WhatsApp code kept (`routers/whatsapp.py` , `models/whatsapp.py` , `wa-service/`).

NFR-18 · Testability and release safety

Current state	172 pytest files (`tests/` , `pytest.ini`). Mandatory pre-release skills: `verify-feature` , `regression-sweep` , `commit-deploy-gate` . Staging at `localhost:8001` before `main` .
Gap	No CI runs the tests (`.github/workflows/` has only deploy and agent build). Release relies on manual discipline. No automatic rollback on failed health check (deliberate, `deploy/oxypc-autodeploy.sh`).

NFR-19 · Data integrity

Current state	Soft delete with `is_trashed` (NOT NULL on main tables), Trash page for restore (`routers/trash.py`). Startup schema validator aborts on unfixable drift (`db_validator.py`). Server-side Decimal money totals (`.claude/skills/server-side-financial-snapshot`). Transaction per request (`database.py`).
Gap	`CLAUDE.md` still says `is_trashed` is nullable; rule and code differ (see 01 §6). No database-level check that soft-deleted rows are excluded from every report.

Open questions

1. What response time and uptime does the business expect (business hours, days, peak user count)?
2. What RPO (data loss) and RTO (time to recover) are acceptable? None are written down.
3. Do security-gate staff use phones for Gate Pass today? Which phone browsers?
4. Is there an accessibility requirement (for example, staff with low vision)?

5. How long must audit logs, backups and uploaded documents be kept (tax/GST rules)?
6. Is moving the LAN app to HTTPS (internal certificate) planned?
7. What is the current production data size (rows in `devices`, `sales`; size of `uploads/`)? The only figure in the repo is "~2 MB" per gzipped dump (`deploy/oxypc-backup.sh`).
8. Which browsers and versions are installed on floor PCs?